Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A motor boat is to reach at a point 30° upstream on other side of a river flowing with the velocity $5\ \mathrm{m/s}$. Velocity of motor boat with respect to water is $5\sqrt{3}\ \mathrm{m/sec}$. The driver should steer the boat at an angle :

(A) 45° w.r.t. the line of destination from starting point.

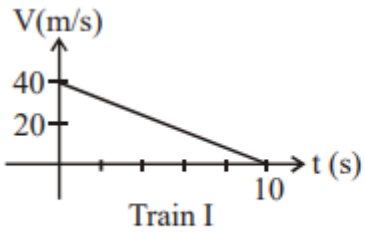
(B) 45° w.r.t. normal to the bank.

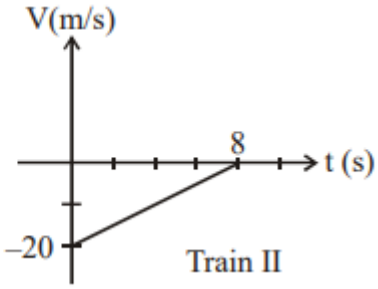
(C) 120° w.r.t. stream direction.

(D) None of these

Q.2

Two trains, which are moving along different tracks in opposite directions, are put on the same track due to a mistake. Their drivers, on noticing the mistake, start slowing down the trains when the trains are $300\ \mathrm{m}$ apart. Graphs given below show their velocities as function of time as the trains slow down, the separation, between the trains when both have stopped, is:





(A) $120\ \mathrm{m}$

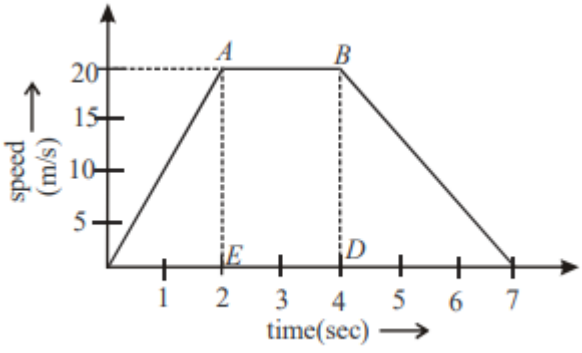
(B) $280\ \mathrm{m}$

(C) $60\ \mathrm{m}$

(D) $20\ \mathrm{m}$

Q.3

Figure gives the speed-time graph of the motion of a car. What is the ratio of the distance travelled by car during the last three seconds to the total distance travelled in seven seconds-



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(A) $1/9$

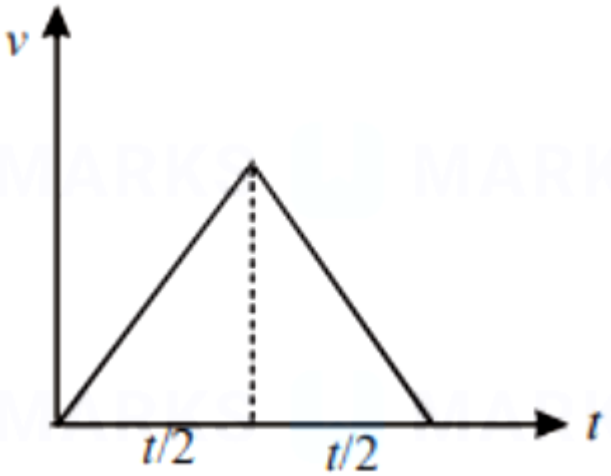
(B) $2/9$

(C) $1/3$

(D) $4/9$

Q.4

The greatest acceleration or deceleration that a train may have is a. The minimum time in which the train may reach from one station to the other separated by a distance d is -



(A) $\sqrt{\frac{d}{a}}$

(B) $\sqrt{\frac{2d}{a}}$

(C) $\frac{1}{2}\sqrt{\frac{d}{a}}$

(D) $2\sqrt{\frac{d}{a}}$

Q.5

A ball is thrown vertically upwards from the ground. It crosses a point at the height of $25\ \mathrm{m}$ twice at an interval of $4\mathrm{sec}$. The ball was thrown with the velocity of -

(A) $20\ \mathrm{m/sec.}$

(B) $25\ \mathrm{m/sec.}$

(C) $30\ \mathrm{m/sec.}$

(D) $35\ \mathrm{m/sec.}$

Q.6

A stone is dropped from a height h. Simultaneously another stone is thrown up from the ground with such a velocity that it can reach a height of $4\ \mathrm{h}$. The two stones cross each other after time-

(A) $\sqrt{\left(\frac{h}{2g}\right)}$

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(B) $\sqrt{\left(\frac{h}{8\,g}\right)}$

(C) $\sqrt{(8hg)}$

(D) $\sqrt{(2hg)}$

Q.7

At the moment $t = 0$, a particle leaves the origin and moves in the positive direction of the x-axis. Its velocity varies with time as
$v = v_0\left(1 - \frac{t}{5}\right)$ where $v_0 = 10\ cm/sec$. The particle will be at the distance of $10\ cm$ from the origin at three different instants T_1, T_2 and $T_3\ [T_3 > T_2 > T_1]$.If the value of $[T_3 - T_2]$ is of the form $\frac{\sqrt{X}-\sqrt{Y}}{2}$ where X and Y are some numbers and $Y < X$. then the value of $[X - Y]/10$ will be ? Assume in this situation negative time is also possible.

(A) 8

(B) 6

(C) 3

(D) 9

Q.8

A point initially at rest moves along x-axis. Its acceleration varies with time as $a = (6t + 5)m/s^2$. If it starts from origin, the distance covered in $2\ s$ is -

(A) $20\ m$

(B) $18\ m$

(C) $16\ m$

(D) $25\ m$

Q.9

Choose the correct options -

(A) A particle moves rectilinearly. Its displacement x at time t is given by $x^2 = at^2 + b$. Its acceleration at time t is proportional to $1/x^3$.

(B) Two bodies begin to free all from same height at a time interval of N sec. If vertical separation between them is 1 after n sec. from the start of I^{st} body, then $n = \frac{1}{gN} + \frac{N}{2}$.

(C) A particle moves rectilinearly. Its displacement x at time t is given by $x^2 = at^2 + b$. Its acceleration at time t is proportional to $1/x^2$.

(D) Two bodies begin to free all from same height at a time interval of $Nsec$. If vertical separation between them is 1 after n sec. from the start of I^{st} body, then $n = \frac{1}{gN} - \frac{N}{3}$.

Q.10

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A particle is projected vertically upwards with a velocity u from a point O. When it returns to the point of projection -

(A) its average velocity is zero

(B) its average speed is $\mathrm{u}/2$

(C) its displacement is zero

(D) its average speed is u.

Q.11

If velocity of the particle is given by $\mathrm{v} = \sqrt{\mathrm{x}}$, where x denotes the positions of the particle and initially particle was at $x = 4$, then which of the following are correct $-$

(A) at $\mathrm{t} = 2\mathrm{sec}$, the position of the particle is at $\mathrm{x} = 9$

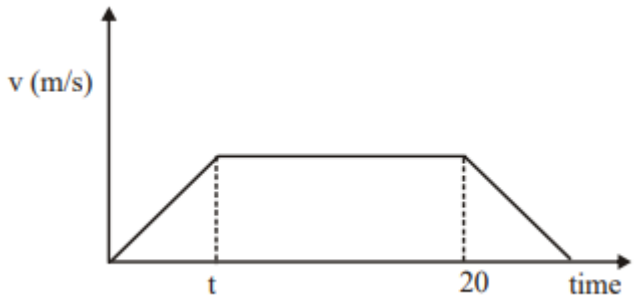
(B) Particle acceleration at $\mathrm{t} = 2\mathrm{sec}$ is $1\ \mathrm{m/s^2}$

(C) Particle acceleration is $1/2\ \mathrm{m/s^2}$ throughout the motion

(D) Particle will never go in negative direction from it's starting position

Q.12

Passage:
The velocity-time graph of a particle moving along a straight line is as shown in figure. The rate of acceleration and deceleration is constant and is equal to $5\ \mathrm{m/s^2}$. If the average velocity during the motion is $20\ \mathrm{m/s}$, then -



Question: The value of t is -

(A) $5\ \mathrm{s}$

(B) $10\ \mathrm{s}$

(C) $20\ \mathrm{s}$

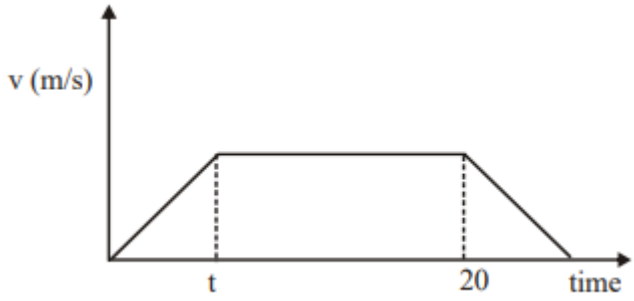
(D) $5\sqrt{2}\ \mathrm{s}$

Q.13

Passage:
The velocity-time graph of a particle moving along a straight line is as shown in figure. The rate of acceleration and

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deceleration is constant and is equal to $5\ \mathrm{m/s^2}$. If the average velocity during the motion is $20\ \mathrm{m/s}$, then -



Question: The maximum velocity of the particle is -

(A) $20\ \mathrm{m/s}$

(B) $25\ \mathrm{m/s}$

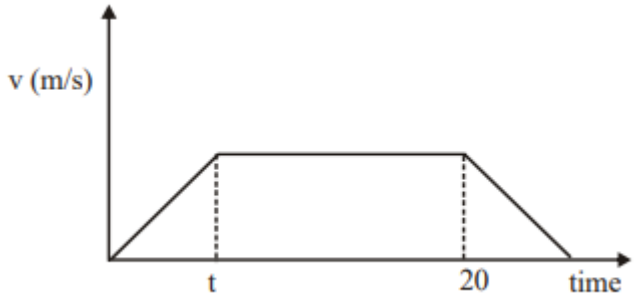
(C) $30\ \mathrm{m/s}$

(D) $40\ \mathrm{m/s}$

Q.14

Passage:

The velocity-time graph of a particle moving along a straight line is as shown in figure. The rate of acceleration and deceleration is constant and is equal to $5\ \mathrm{m/s^2}$. If the average velocity during the motion is $20\ \mathrm{m/s}$, then -



Question: The distance travelled with uniform velocity is -

(A) $375\ \mathrm{m}$

(B) $125\ \mathrm{m}$

(C) $300\ \mathrm{m}$

(D) $450\ \mathrm{m}$

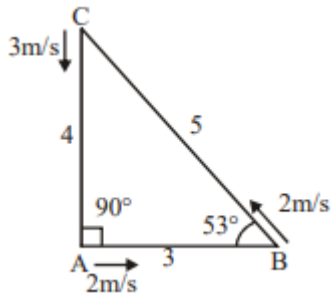
Q.15

Passage:

3 solid rods are joined together to form a triangle ABC.3 ants a, b and c initially situated at A, B and C respectively start moving in directions along the rods AB, BC and CA respectively at time $\mathrm{t} = 0$, with constant

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speeds as shown in figure.



After reaching any point among A, B or C they continue their motion in the anticlockwise direction along the sides of the triangle. Neglect any collision between ants.

Question: Find the velocity of approach of a and b at $t = 0$ sec.

(A) $4\ m/s$

(B) $3.6\ m/s$

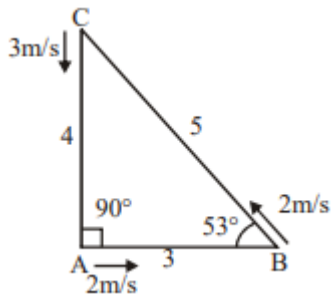
(C) $3.2\ m/s$

(D) None of these

Q.16

Passage:

3 solid rods are joined together to form a triangle $ABC.3$ ants a, b and c initially situated at A, B and C respectively start moving in directions along the rods AB, BC and CA respectively at time $t = 0$, with constant speeds as shown in figure.



After reaching any point among A, B or C they continue their motion in the anticlockwise direction along the sides of the triangle. Neglect any collision between ants.

Question: What will be the minimum distance between ant a and b during the motion?

(A) $3\ m$

(B) $\frac{3}{\sqrt{2}}\ m$

(C) $0.9\ m$

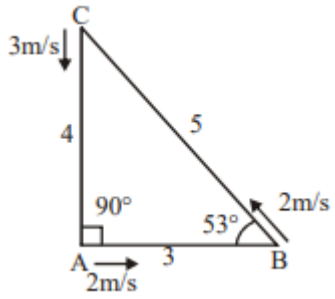
(D) $1.7\ m$

Q.17

Passage:

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3 solid rods are joined together to form a triangle ABC.3 ants a, b and c initially situated at A, B and C respectively start moving in directions along the rods AB, BC and CA respectively at time $t = 0$, with constant speeds as shown in figure.



After reaching any point among A, B or C they continue their motion in the anticlockwise direction along the sides of the triangle. Neglect any collision between ants.

Question: At what time t the distance between a and c is minimum for the first time.

(A) $t = 0$ sec.

(B) $t = 4$ sec.

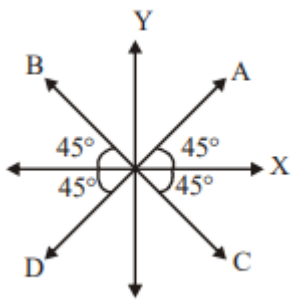
(C) $t = 2$ sec.

(D) None of these

Q.18

Consider the velocity vectors of four particles A, B, C and D as shown in figure. Speed of the particles is same. Take $+x$-axis in east and $+y$-axis in north direction.

Column I	**Column II**
(A) The particle(s) is/are going in north direction with respect to atleast one of the other particle.	(p) A
(B) The particle(s) is/are going in south direction with respect to atleast one of the other particle.	(q) B
(C) The particle(s) is/are going in east direction with respect to atleast one of the other particle.	(r) C
(D) The particle(s) is/are going in west direction with respect to atleast one of the other particle.	(s) D (t) None



(A) $A - (pq)B - (rs)C - (pr)D - (qs)$

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(B) $A-(rs)B-(pq)C-(qs)D-(pr)$

(C) $A-(qs)B-(rs)C-(pq)D-(pr)$

(D) $A-(pqs)B-(pr)C-(qs)D-(ps)$

Q.19

An airplane pilot sets a compass course due west and moves with a constant speed $12\ \mathrm{km/hr}$ with respect to air. After flying for $1/2$ hour, he finds himself over a town that is $7.5\ \mathrm{km}$ west and $2\ \mathrm{km}$ south of his starting point. Find the magnitude of wind velocity (in $\mathrm{km/hr}$).

Q.20

Inside a starship at rest on the earth, a ball rolls off the top of a horizontal table and lands a distance D from the foot of the table. This starship now lands on the unexplored Planet X. The commander, Captain Curious, rolls the same ball off the same table with the same initial speed as on earth and finds that it lands a distance $2.76\ D$ from the foot of the table. What is the acceleration due to gravity on Planet X ?

Answers & Solutions

Q.1 Answer:

120° w.r.t. stream direction.

Solution:

The velocity of motor boat is given

As $\vec{v}_{m}=\vec{v}_{mw}+\vec{v}_{w} \Rightarrow \vec{v}_{m}=-(5\sqrt{3}\cos 30)\hat{i}+(5\sqrt{3}\cos 30)\hat{j}+5\hat{i}$

$\Rightarrow \quad \vec{v}_{m}=-7.5\mathrm{i}+\frac{5\sqrt{3}}{2}\hat{j}+5\hat{i}$

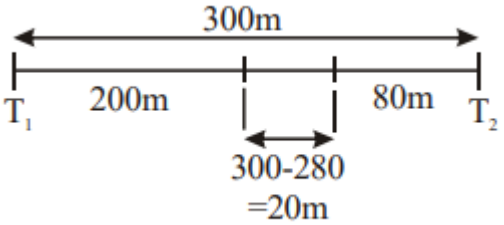
$\phi=\tan^{-1}\left[-\frac{5\sqrt{3}}{2\times 2.5}\right]=\tan^{-1}(-\sqrt{3})=120^{\circ}$

Q.2 Answer:

$20\ \mathrm{m}$

Solution:

Initial distance between trains is $300\ \mathrm{m}$**. Displacement of 1 st train is calculated by area under V-t.**



Displacement of train $1=\frac{1}{2}\times 10\times 40=200\ \mathrm{m}$

Displacement of train $2=\frac{1}{2}\times 8\times(-20)=-80\ \mathrm{m}$

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Which means it moves towards left.

$\therefore$ **Distance between the two is** $20\ \mathrm{m}$.

Q.3 Answer:

$1/3$

Solution:

Distance travelled in last three second $= \frac{1}{2} \times 3 \times 20 = 30\ \mathrm{m}$

Distance travelled in seven second $= \frac{1}{2} \times (2+7) \times 20 = 90\ \mathrm{m}$

[Trapezium area $= (\mathrm{a}+\mathrm{b})\mathrm{d}$ **]**

Ratio $= \frac{30}{90} = \frac{1}{3}$

Q.4 Answer:

$2\sqrt{\frac{\mathrm{d}}{\mathrm{a}}}$

Solution:

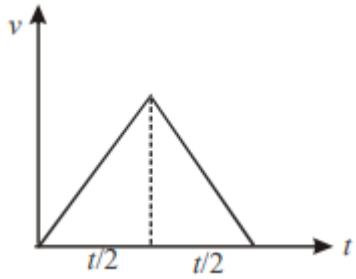
$\mathrm{v} = 0 + \frac{\mathrm{at}}{2}$,

Area under $\mathrm{v} - \mathrm{t}$ **graph**

$\mathrm{d} = \frac{1}{2}\mathrm{tv}$

For minimum time $\mathrm{d} = \frac{1}{2}\frac{\mathrm{aa}}{(\mathrm{a}+\mathrm{a})}\mathrm{t}^2$;

$\mathrm{t} = 2\sqrt{\frac{\mathrm{d}}{\mathrm{a}}}$



Q.5 Answer:

$30\ \mathrm{m/sec.}$

Solution:

$h = ut - \frac{1}{2}gt^2$

Difference of root : $\mathrm{t}_2 - \mathrm{t}_1 = \frac{2\sqrt{\mathrm{u}^2 - 2\mathrm{gh}}}{\mathrm{g}}$

$\mathrm{t}_2 = \frac{+\mathrm{u} + \sqrt{\mathrm{u}^2 - 4 \times 1/2\ \mathrm{g}(+\mathrm{h})}}{2 \times 1/2\ \mathrm{g}} = \frac{\mathrm{u} + \sqrt{\mathrm{u}^2 - 2\mathrm{gh}}}{\mathrm{g}}$

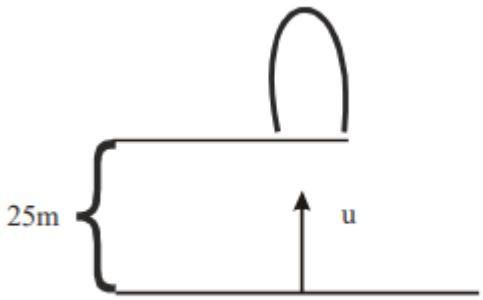
$\mathrm{t}_1 = \frac{\mathrm{u} - \sqrt{\mathrm{u}^2 - 2\mathrm{gh}}}{\mathrm{g}}$;

$\mathrm{t}_2 - \mathrm{t}_1 = 4$,

$4 = \frac{2\sqrt{\mathrm{u}^2 - 2\ \mathrm{g} \times 25}}{\mathrm{g}}; \mathrm{g} = 10\ \mathrm{m/s}^2$,

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\mathrm{u}=30 \mathrm{~m} / \mathrm{s}$



Q.6 Answer:

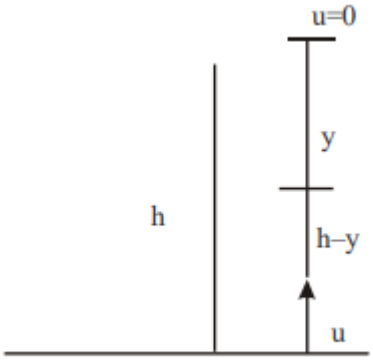
$\sqrt{\left(\frac{\mathrm{h}}{8\,\mathrm{g}}\right)}$

Solution:

$4\,\mathrm{h}=\frac{\mathrm{u}^2}{2\,\mathrm{g}} \Rightarrow \mathrm{u}=\sqrt{8\mathrm{gh}}$

$\mathrm{y}=\frac{1}{2}\mathrm{gt}^2 ; \mathrm{h}-\mathrm{y}=\mathrm{ut}-\frac{1}{2}\mathrm{gt}^2$

$\mathrm{h}=\mathrm{ut} \Rightarrow \mathrm{t}=\frac{\mathrm{h}}{\sqrt{8\mathrm{gh}}}=\sqrt{\frac{\mathrm{h}}{8\,\mathrm{g}}}$



Q.7 Answer:

8

Solution:

$$\frac{dx}{dt}=v_0-\frac{v_0 t}{5}$$

$$\Rightarrow x=v_0 t-\frac{v_0 t^2}{10}$$

where x can be either $+10$ **or** -10**.**

$$10t-t^2=10 \Rightarrow t=\frac{10 \pm \sqrt{60}}{2}$$

$$10t-t^2=-10 \Rightarrow t=\frac{10 \pm \sqrt{140}}{2}$$

for 2nd **event.** $t=\frac{10-\sqrt{60}}{2}$

and for 3rd **event,** $t=\frac{10-\sqrt{140}}{2}$

$\frac{(140-60)}{10}=8$

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Q.8 Answer:

$18\ \mathrm{m}$

Solution:

Given $a = \frac{dv}{dt} = 6t + 5$ **or** $dv = (6t + 5)dt$

Integrating, we get $\int_0^v dv = \int_0^t (6t + 5)dt$ **or** $v = \left(\frac{6t^2}{2} + 5t\right)$

Again $v = \frac{ds}{dt} \quad \therefore ds = \left(\frac{6t^2}{2} + 5t\right) dt$

Integrating again, we get $\int_0^s ds = \int_0^t \left(\frac{6t^2}{2} + 5t\right) dt$

$\therefore \quad s = \frac{3t^3}{3} + \frac{5t^2}{2}$

When, $t = 2\ s,\ s = 3 \times \frac{2^3}{3} + \frac{5 \times 2^2}{2} = 3 \times \frac{8}{3} + \frac{5 \times 4}{2} = 8 + 10 = 18\ m$

Q.9 Answer:

A particle moves rectilinearly. Its displacement x at time t is given by $x^2 = at^2 + b$. Its acceleration at time t is proportional to $1/x^3$.

Solution:

(A) $x^2 = at^2 + b$

$2x\frac{dx}{dt} = 2at + 0$

$xv = at$

$x\frac{dv}{dt} + v\frac{dx}{dt} = a; x\frac{dv}{dt} = a - v^2$

$\frac{dv}{dt} = \frac{a - v^2}{x} = \frac{a - \frac{a^2t^2}{x^2}}{x} = \frac{ab}{x^3}; \frac{dv}{dt} \propto \frac{1}{x^3}$

(B) $S_1 = \frac{1}{2}gn^2,\ S_2 = \frac{1}{2}\ g(n - N)^2$

$S_1 - S_2 = \frac{1}{2}\ g\left[n^2 - (n - N)^2\right]; 1 = \frac{g}{2}(2n - N)N; n = \frac{1}{gN} + \frac{N}{2}$

Q.10 Answer:

its average velocity is zero

Solution:

(A) Total displacement is zero hence its average velocity is zero.

(B) Displacement is zero

(C) Total distance travelled is $2\ s$ **and total time taken is** $2t$**.**

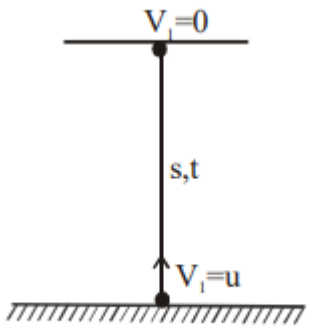
$0^2 = u^2 - 2gs$

$\therefore s = \frac{u^2}{2g} \Rightarrow 2\ s = \frac{u^2}{g}$

Also, $\quad 0 = u - gt \Rightarrow t = u/g$

$\therefore \quad 2t = \frac{2u}{g}; <\ \text{speed}\ > = \frac{u^2/g}{2u/g} = \frac{u}{2}$

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Q.11 Answer:

at $\mathrm{t} = 2\mathrm{sec}$, the position of the particle is at $\mathrm{x} = 9$

Solution:

$v = \sqrt{x} \Rightarrow \frac{dx}{dt} = \sqrt{x}$

$\frac{\mathrm{dx}}{\mathrm{x}^{1/2}} = \mathrm{dt} \Rightarrow 2\sqrt{\mathrm{x}} = \mathrm{t} + \mathrm{C}$

but given at $\mathrm{t} = 0, \mathrm{x} = 4 \Rightarrow \mathrm{c} = 4$

$x = \frac{(t+4)^2}{4} \Rightarrow x = \frac{(6)^2}{4} = \frac{36}{4} = 9m$ (**Putting** $t = 2\mathrm{sec}$)

$\mathrm{a} = \mathrm{v}\frac{\mathrm{dv}}{\mathrm{dx}} = \sqrt{\mathrm{x}} + \frac{1}{2\sqrt{\mathrm{x}}} = \frac{1}{2}\ \mathrm{m/s^2}$

Q.12 Answer:

$5\ \mathrm{s}$

Solution:

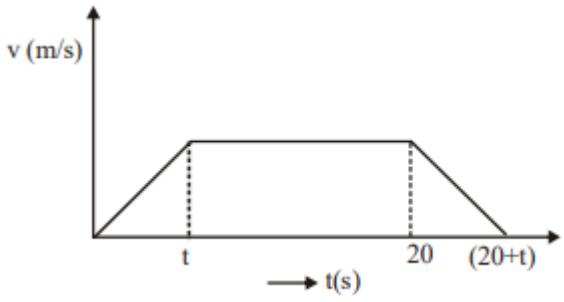
(i) Average speed $= 20\ \mathrm{m/s}$

$$\frac{\text{Total distance}}{\text{Total time}} = 20 \Rightarrow \frac{\text{Area of graph}}{20 + \mathrm{t}} = 20$$

$$\Rightarrow \frac{1}{2}(20 + \mathrm{t} + 20 - \mathrm{t}) \times 5\mathrm{t} = 20(20 + \mathrm{t}) \Rightarrow \mathrm{t} = 5\ \mathrm{s}$$

(ii) **Maximum velocity** $= 5\mathrm{t} = 5 \times 5 = 25\ \mathrm{m/s}$

(iii) $5\mathrm{t}(20 - \mathrm{t}) = 100\mathrm{t} - 5\mathrm{t}^2 = 375\ \mathrm{m}$



Q.13 Answer:

$25\ \mathrm{m/s}$

Solution:

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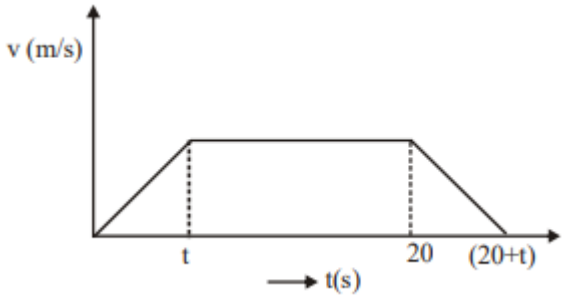
(i) Average speed $= 20\ \mathrm{m/s}$

$$\frac{\text{Total distance}}{\text{Total time}} = 20 \Rightarrow \frac{\text{Area of graph}}{20+t} = 20$$

$$\Rightarrow \frac{1}{2}(20+t+20-t)\times 5t = 20(20+t) \Rightarrow t = 5\ \mathrm{s}$$

(ii) Maximum velocity $= 5t = 5\times 5 = 25\ \mathrm{m/s}$

(iii) $5t(20-t) = 100t - 5t^2 = 375\ \mathrm{m}$



Q.14 Answer:

$375\ \mathrm{m}$

Solution:

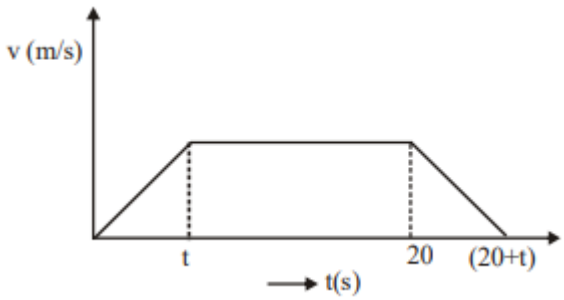
(i) Average speed $= 20\ \mathrm{m/s}$

$$\frac{\text{Total distance}}{\text{Total time}} = 20 \Rightarrow \frac{\text{Area of graph}}{20+t} = 20$$

$$\Rightarrow \frac{1}{2}(20+t+20-t)\times 5t = 20(20+t) \Rightarrow t = 5\ \mathrm{s}$$

(ii) Maximum velocity $= 5t = 5\times 5 = 25\ \mathrm{m/s}$

(iii) $5t(20-t) = 100t - 5t^2 = 375\ \mathrm{m}$



Q.15 Answer:

$3.2\ \mathrm{m/s}$

Solution:

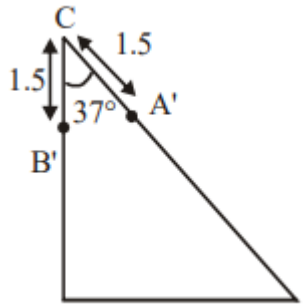
$v_{\text{app}} = 2 + 2\cos 53^\circ = 3.2\ \mathrm{m/s}$

Position of a and b at minimum distance by cosine law $\mathrm{A'B'}^2 = (1.5)^2 + (1.5)^2 - 2\times(1.5)^2\cos 37^\circ$

$\mathrm{A'B} = 0.9\ \mathrm{m}$

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Minimum distance between a and c is zero when they meet first time $t = \frac{\ell_{rel}}{v_{rel}} = \frac{4}{1} = 4\text{sec.}$



Q.16 Answer:

$0.9\ \text{m}$

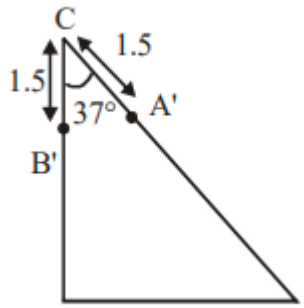
Solution:

$v_{app} = 2 + 2\cos 53^\circ = 3.2\ \text{m/s}$

Position of a and b at minimum distance by cosine law $A'B'^2 = (1.5)^2 + (1.5)^2 - 2 \times (1.5)^2 \cos 37^\circ$

$A'B = 0.9\ \text{m}$

Minimum distance between a and c is zero when they meet first time $t = \frac{\ell_{rel}}{v_{rel}} = \frac{4}{1} = 4\text{sec.}$



Q.17 Answer:

$t = 4\ \text{sec.}$

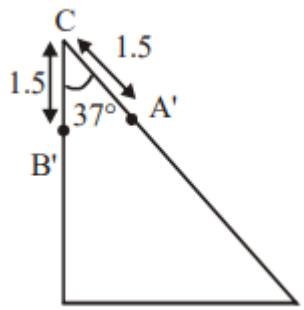
Solution:

$v_{app} = 2 + 2\cos 53^\circ = 3.2\ \text{m/s}$

Position of a and b at minimum distance by cosine law $A'B'^2 = (1.5)^2 + (1.5)^2 - 2 \times (1.5)^2 \cos 37^\circ$

$A'B = 0.9\ \text{m}$

Minimum distance between a and c is zero when they meet first time $t = \frac{\ell_{rel}}{v_{rel}} = \frac{4}{1} = 4\text{sec.}$



Q.18 Answer:

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$A-(pq)B-(rs)C-(pr)D-(qs)$

Solution:

$(A)-pq,(B)-rs,(C)-pr,(D)-qs$

$\overrightarrow{\mathrm{V}}_{\mathrm{A}}=\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{i}}+\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{j}};\overrightarrow{\mathrm{V}}_{\mathrm{B}}=-\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{i}}+\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{j}};\overrightarrow{\mathrm{V}}_{\mathrm{C}}=\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{i}}-\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{j}};\overrightarrow{\mathrm{V}}_{\mathrm{D}}=-\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{i}}-\frac{\mathrm{V}}{\sqrt{2}}\hat{\mathrm{j}}$

Q.19 Answer:

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Solution:

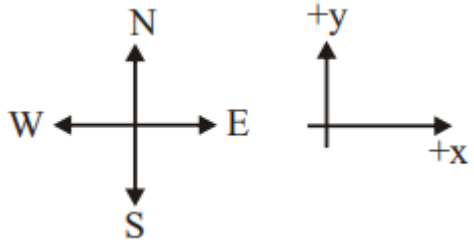
$\mathrm{u}_{\text{plane}}-\mathrm{u}_{\text{air}}=-12\hat{\mathrm{i}}$

If $u_{\text{air}}=x\hat{i}+y\hat{j};\quad u_{\text{plane}}=(x-12)\hat{i}+y\hat{j}$

$\mathrm{S}=\mathrm{ut}\Rightarrow\frac{\mathrm{x}-12}{2}=-7.5$ **and** $\frac{\mathrm{y}}{2}=-2$

$\Rightarrow\quad x=-3\text{kmph}$ **and** $y=-4\text{kmph}$

Ans. 5kmph



Q.20 Answer:

—

Solution:

IDENTIFY: The ball moves with projectile motion with an initial velocity that is horizontal and has magnitude v_0.

The height h **of the table and** v_0 **are the same; the acceleration due to gravity changes from** $g_{\mathrm{E}}=9.80\ \mathrm{m/s^2}$ **on earth to** g_X **on planet** X.

SET UP: Let $+x$ **be horizontal and in the direction of the initial velocity of the marble and let** $+y$ **be upward.**

$v_{0x}=v_0,v_{0y}=0,a_x=0,a_y=-\mathrm{g}$, **where** g **is either** g_{E} **or** g_{X}.

EXECUTE: Use the vertical motion to find the time in the air: $y-y_0=-h.\quad y=y_0=v_0t+\frac{1}{2}a_yt^2$ **gives** $t=\sqrt{\frac{2h}{g}}$.

Then $x-x_0=v_{0,x}t+\frac{1}{2}a_xt^2$ **gives** $x-x_0=v_{0,x}t=v_0\sqrt{\frac{2h}{g}},x-x_0=D$ **on earth and** $2.76D$ **on Planet X.**

$(x-x_0)\sqrt{g}=v_0\sqrt{2h}$, **which is constant, so** $D\sqrt{g_{\mathrm{E}}}=2.76D\sqrt{g_{\mathrm{X}}}.\ g_{\mathrm{X}}=\frac{g_{\mathrm{E}}}{(2.76)^2}=0.131g_{\mathrm{E}}=1.28\ \mathrm{m/s^2}$.

EvALUATE: On Planet X **the acceleration due to gravity is less, it takes the ball longer to reach the floor and it travels farther horizontally.**

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